## 3.3.9 Material management

### Cut materials efficiently and minimise waste

| Content | Potential links to maths and science |
|---|---|
| The importance of planning the cutting and shaping of material to minimise waste eg nesting of shapes and parts to be cut from material stock forms.<br><br>How additional material may be removed by a cutting method or required for seam allowance, joint overlap etc. | Expression in decimal and standard form eg calculation of required materials.<br><br>Calculate surface area and volume eg material requirements.<br><br>Angular measures eg measurement and marking out.<br><br>SI units eg measurement of materials and components using standard units as appropriate.<br><br>The use of reference datum points and co-ordinates. |

### Use appropriate marking out methods, data points and coordinates

| Content | Potential links to maths and science |
|---|---|
| The value of using measurement and marking out to create an accurate and quality prototype.<br><br>The use of data points and coordinates including the use of reference points, lines and surfaces, templates, jigs and/or patterns | Use angular measures eg tessellation of component parts.<br><br>Calculating material area eg working out the quantity of materials required.<br><br>SI units eg accurate use of appropriate units of measurement to calculate material requirements. |